E-Commerce Sales Data Analysis Using Python and Tableau

Project Documentation

Name: VISHAL A

Course: B.Tech- Computer Science and Business Systems

Institution: Rajalakshmi Engineering College

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1. Introduction:

E-commerce platforms generate large volumes of transactional data every day. Analyzing this data helps businesses understand customer purchasing behavior, identify sales trends, and make informed strategic decisions. This project focuses on analyzing an e-commerce dataset to identify patterns in sales performance, customer locations, product categories, and payment methods. Python was used for data cleaning and analysis, while Tableau was used to build an interactive dashboard that visually represents the insights extracted from the data.

2. Objective of the Project:

The objective of this project is to analyze e-commerce sales data and extract meaningful insights that can help businesses understand their performance and customer behavior. The project aims to identify monthly sales trends, determine top-performing cities, analyze product category performance, examine customer payment preferences, and visualize the results through an interactive dashboard.

3. Tools and Technologies Used:

The following tools and technologies were used to complete this project:
• Python – Data processing and analysis.
• Pandas – Data manipulation and cleaning.
• NumPy – Numerical computations.
• Google Colab / Jupyter Notebook – Running and testing Python code.
• Tableau – Data visualization and dashboard creation.

4. Dataset Description:

The dataset used in this project contains transactional data from an e-commerce platform. It includes information related to customers, orders, products, sellers, and payments. Important fields in the dataset include order identifiers, purchase timestamps, product categories, payment types, payment values, and customer locations. This dataset provides a comprehensive view of the sales process, from order placement to delivery.

5. Data Analysis Process:

The project followed a structured data analysis workflow. First, the dataset was loaded into Python using the Pandas library. Next, the data was cleaned by handling missing values and ensuring that the columns were formatted correctly. After cleaning, different tables were merged to create a unified dataset for analysis. Additional analytical fields

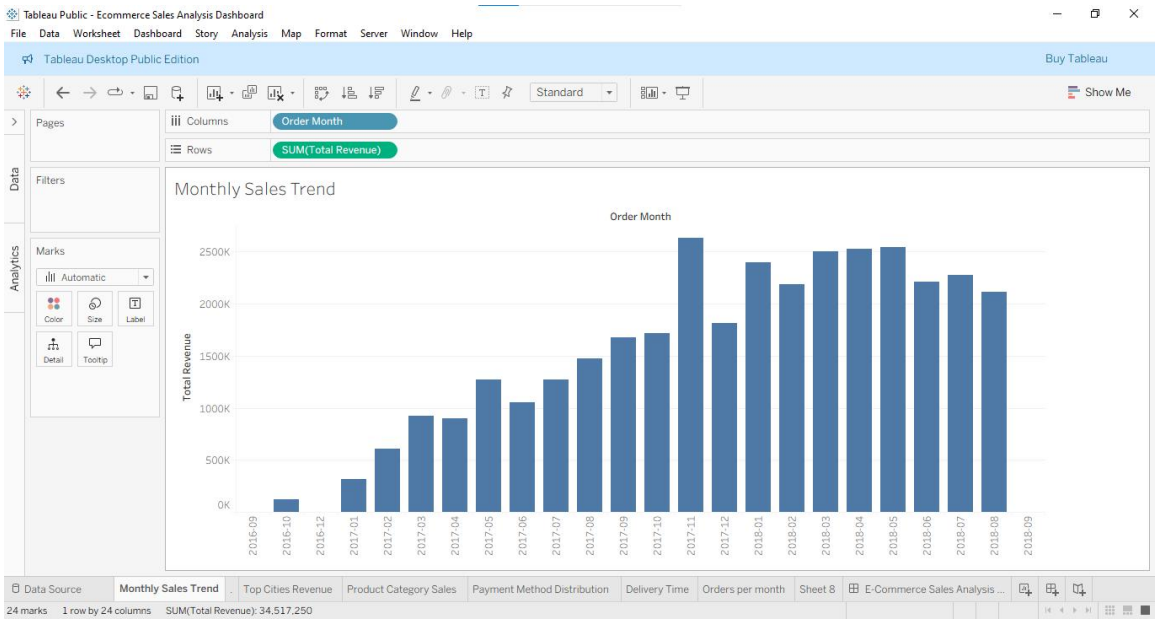
such as total revenue and monthly sales indicators were created. Finally, the processed dataset was exported and used in Tableau to build visual dashboards.

6. Dashboard Visualizations:

The following visualizations were created in the Tableau dashboard to represent the insights obtained from the data analysis.

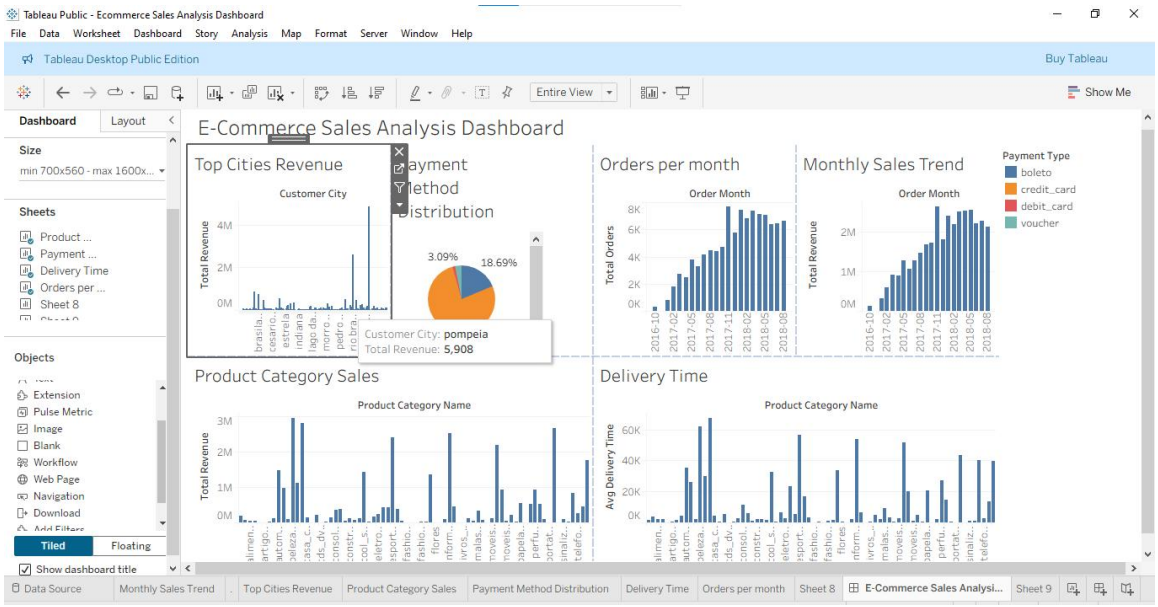
Monthly Sales Trend

Description: Explain briefly what this chart shows and what insight it provides.



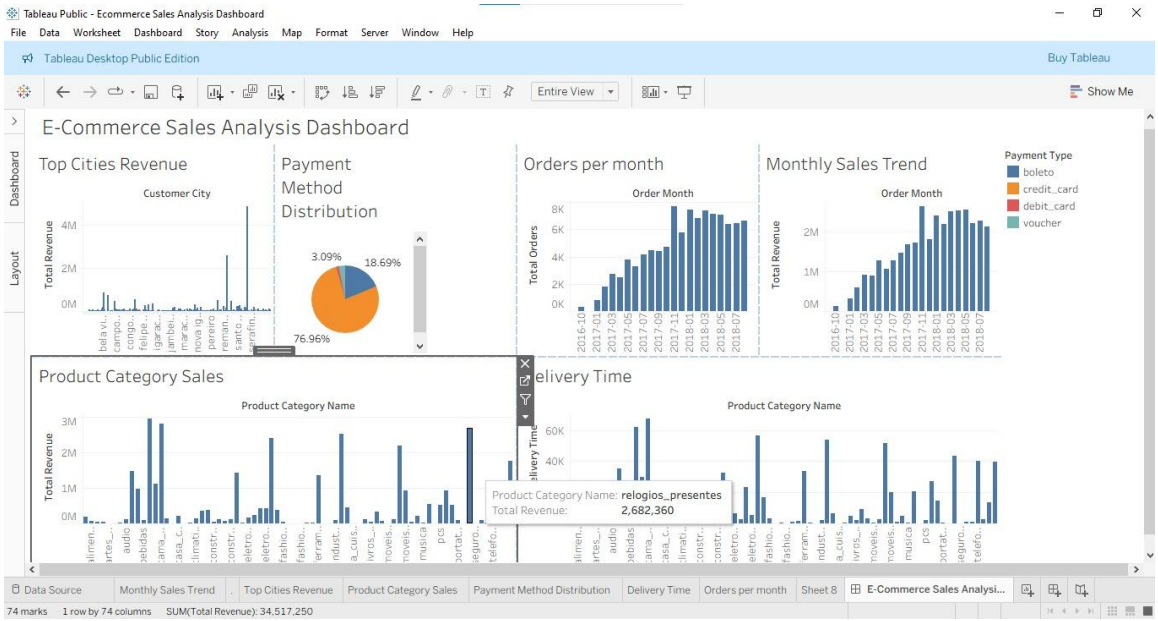
Top Cities by Revenue

Description: Explain briefly what this chart shows and what insight it provides.



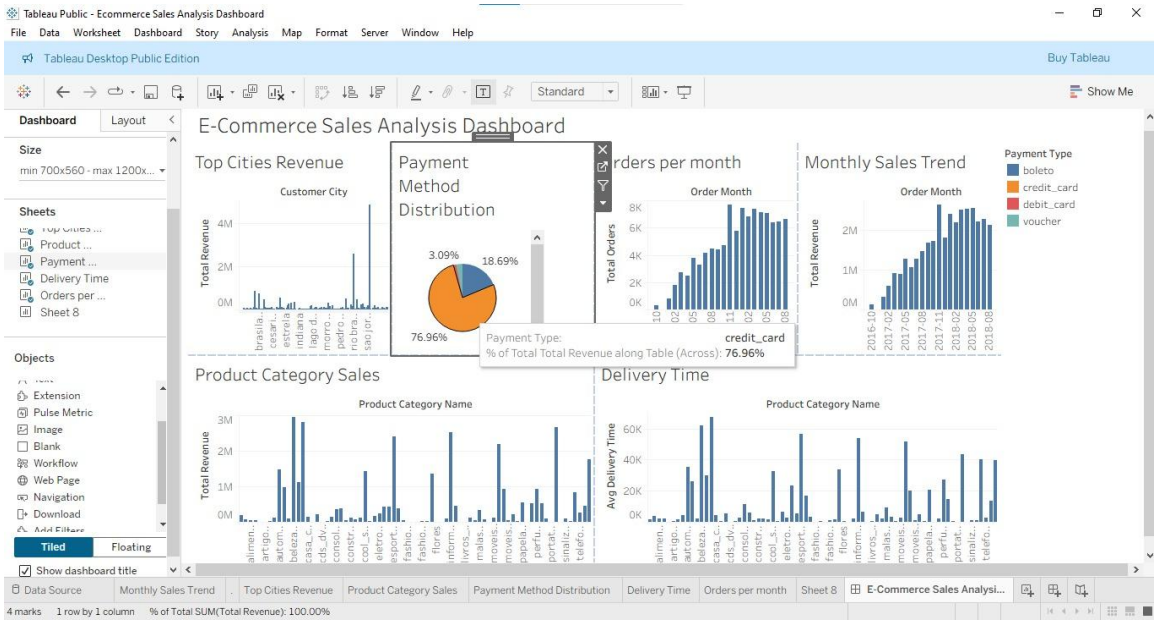
Product Category Sales

Description: Explain briefly what this chart shows and what insight it provides.



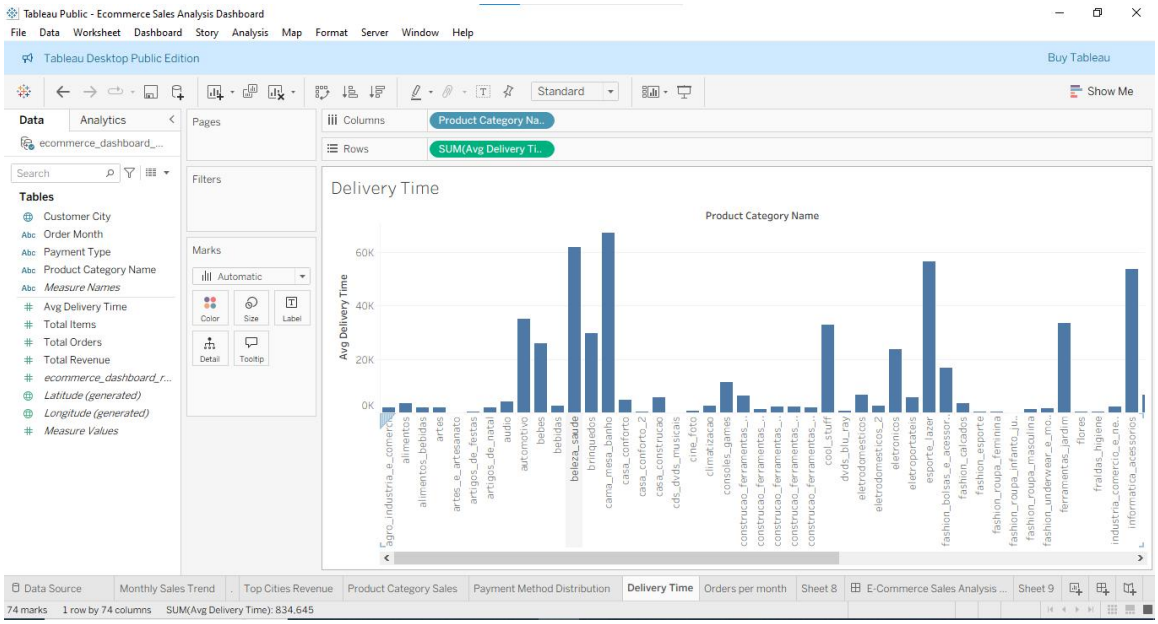
Payment Method Distribution

Description: Explain briefly what this chart shows and what insight it provides.



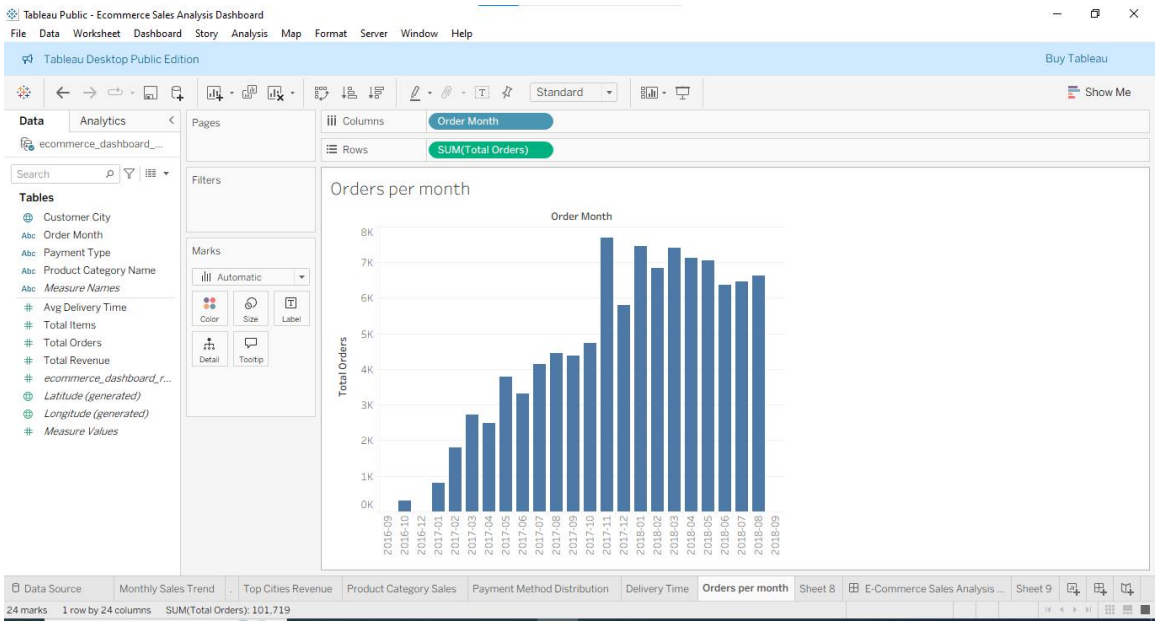
Delivery Time Analysis

Description: Explain briefly what this chart shows and what insight it provides.

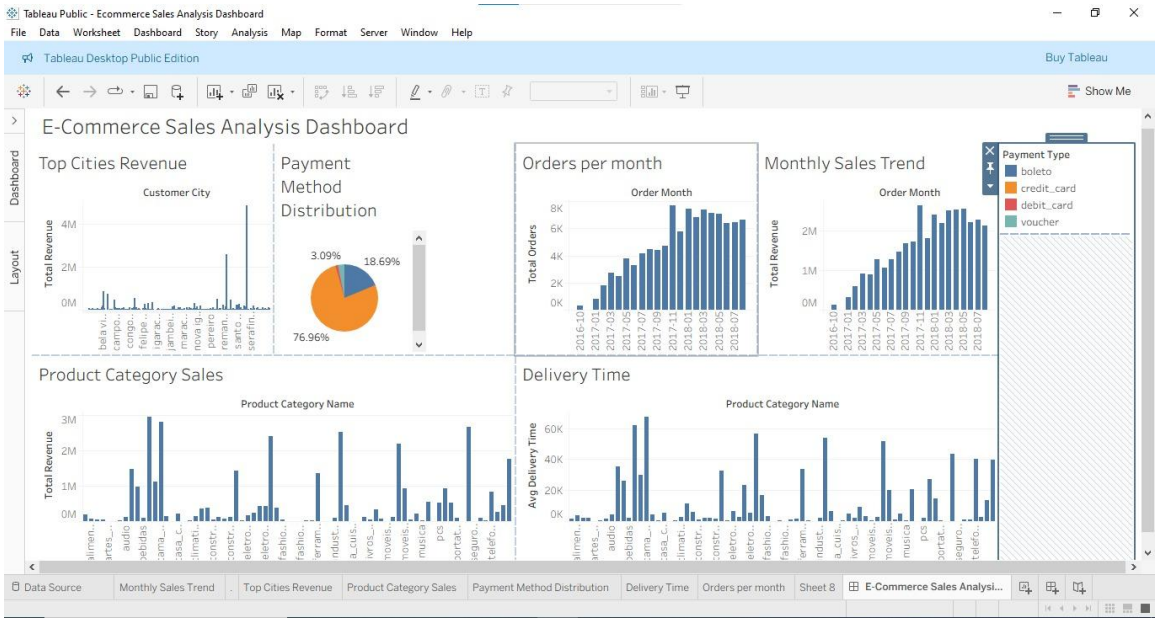


Orders per Month

Description: Explain briefly what this chart shows and what insight it provides.



Final Dashboard



7. Key Insights:

Several insights were identified during the analysis. Credit card payments represent the majority of transactions. Certain cities contribute significantly to total revenue. Some product categories generate higher sales compared to others. The monthly sales trend highlights periods of higher customer activity. These insights help businesses better understand customer preferences and improve decision-making processes.

8. Conclusion:

This project demonstrates how data analysis techniques can be used to extract meaningful information from e-commerce data. By combining Python for data preprocessing and Tableau for visualization, it is possible to uncover patterns in sales performance, customer preferences, and operational efficiency. The interactive dashboard created in this project provides a clear and concise overview of business performance.

9. Future Improvements:

Future work could extend this project by incorporating predictive analytics techniques such as sales forecasting using machine learning models. Additional analyses such as customer segmentation and product recommendation systems could also be implemented to provide deeper business insights.

10. Author / Project Information:

Name: VISHAL A

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Project Description:

This project analyses an e-commerce dataset to identify sales trends, customer behaviour, product performance, and payment patterns. Python was used for data cleaning and analysis, and Tableau was used to create interactive dashboards to visualize the insights.

Date of Completion: 12-03-2026.